WoMAT — Evaluation Plan

Experiments and graphs for voxelization, voxel size, centrality, neighbour approximation, and safety with movement

Master's Thesis Technical Note

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Purpose

This note proposes experiments and graphs for the evaluation chapter of the thesis. Every component that the thesis contributes is covered by at least one experiment: the voxelization of shared workspaces, the automatic voxel size estimation, the centrality-based selection of handover points, the cheap neighbour approximation, and the safety and movement concept. An end-to-end evaluation ties the components together. For each experiment, the question it answers, the metrics, the proposed graph, and the data that already exists are listed. Nothing has been implemented yet.

Existing data

Part of the required data is already recorded by the running system:

- `performance_log/concurrency.jsonl`, `live.jsonl`: motion requests, start and refusal events, wait times, claimed voxels, contested regions, and the number of concurrently moving arms, for the *region* and *exclusive* arbitration modes.
- `performance_log/observed-region.jsonl`, `observed-exclusive.jsonl`: joint traces of all robots, sampled every 100 ms.
- `performance_log/interaction-compute.jsonl`: per-pair interaction-space computations with voxel count, duration, and cache status.
- `docs/voxel_grid_size_approximation.md`, Section 13: an already specified evaluation of the voxel size estimator (baselines, metrics, sensitivity study).

Prerequisite: reproducibility

The RobWoT workspace sampler never seeds `math.random` (`robot_driver_workspace.lua`, line 150), so repeated runs produce different workspaces. Before any experiment is run, a seed parameter should be added. All experiments should then be repeated over about ten seeds, and the dispersion should be reported instead of single values.

1 Voxelization of the shared workspace

E1 Convergence with sample count

- **Question:** How many workspace samples are needed until the shared region stops changing?

- **Setup:** Sample each workspace with 5k, 10k, 25k, 50k, and 100k joint configurations.
- **Metrics:** shared-voxel count and shared volume per robot pair; computation time.
- **Graph:** shared volume over sample count (logarithmic x-axis), one line per robot pair, with a band showing the dispersion over seeds.

E2 Per-pair overview

- **Setup:** All eight robot pairs of the five-robot cell.
- **Metrics:** voxel count, volume, computation time, cache hit.
- **Data:** largely available in `interaction-compute.jsonl`.
- **Graph:** table, complemented by two-dimensional projections (top view) of the shared regions for two or three representative pairs, including a degenerate pair such as Robot 1–Robot 3 with only 2–3 voxels.

2 Voxel size estimation

This part follows the plan in Section 13 of `voxel_grid_size_approximation.md`.

E3 Accuracy against baselines

- **Baselines:** fixed voxel size of 0.05 m; bounding-box-only estimate; the proposed bidirectional proximity estimator; an offline multi-resolution sweep, used only as a reference.
- **Metrics:** absolute and relative error from N_{target}; estimator time and one-shot voxelization time; Jaccard similarity $J(A, B) = |A \cap B|/|A \cup B|$ of the interaction cells against a fine-grid reference.
- **Graph:** estimated over actual voxel count per pair on log–log axes, with a $y = x$ reference line and one marker per method.

E4 Accuracy–cost trade-off

- **Setup:** Sweep the voxel size from about 0.01 m to 0.1 m.
- **Metrics:** Jaccard similarity to the reference; computation time.
- **Graph:** Pareto-style curve of Jaccard similarity over computation time, with the voxel size chosen by the estimator marked. This is the strongest argument that the automatic choice is sensible.

E5 Sensitivity

- **Factors:** $N_{\text{target}} \in \{250, 500, 1000, 2000, 4000\}$, $q \in \{0.50, 0.75, 0.90\}$, $\kappa \in \{1.0, 1.25, 1.5\}$.
- **Metrics:** resulting voxel size and voxel count, over repeated point-cloud generations.
- **Graph:** small multiples of heatmaps, or box plots of the final voxel size across seeds.

3 Centrality: handover point selection

E6 Selection strategies

- **Strategies:** current score (centrality combined with density); nearest to the centroid only; density only; random shared voxel; bounding-box centre.
- **Metrics:** rate at which both robots find a valid joint solution for the point; number of refused candidates before acceptance; distance to the boundary of the shared region; edge cost (TCP distance).
- **Graph:** bar chart of the feasibility rate per strategy; histogram of refused candidates; two-dimensional heatmap slice of the scores with the selected point marked.

E7 Robustness to calibration error

- **Setup:** Perturb one robot base by ±1–2 cm and repeat the selection.
- **Metrics:** whether the selected point remains feasible for both robots.
- **Graph:** success rate over perturbation magnitude, one line per strategy.

4 Cheap neighbour approximation

E8 Accuracy against ground truth

- **Setup:** All robot pairs of the two- to five-robot scenes; the reach-sphere test is compared with the actual shared-voxel count.
- **Metrics:** true and false positives and negatives, using a minimum-voxel threshold (for example, at least 10 shared voxels) as the definition of a real neighbour.
- **Graph:** scatter plot of base distance over combined reach, coloured by shared-voxel count, with the decision boundary $d = r_1 + r_2$. False positives such as Robot 1–Robot 3 become visible immediately. Complemented by a confusion matrix.

E9 Time saved

- **Metrics:** microseconds per reach-sphere test against seconds per full interaction-space computation; number of computations avoided.
- **Graph:** number of computed pairs over the number of robots, with and without the filter ($\mathcal{O}(n^2)$ against the number of actual neighbours).

5 Safety and movement

E10 Cost of collision checking

- **Setup:** Axis-aligned bounding-box broad phase against exhaustive checking of all shape pairs.
- **Metrics:** time per collision check; time over the number of shape pairs. Earlier measurements indicated about 26 ms per full check and a speed-up of roughly 13× with the broad phase; these values are not stored in the repository and have to be measured again.
- **Graph:** bar chart of the time per check; line plot of time over the number of shape pairs.

E11 Pipeline test course and ablation

- **Setup:** 100–200 random targets around the obstacles; the pipeline is run completely and with individual stages disabled.

- **Metrics:** outcome per target: direct success, success via OMPL detour, rejected before motion (workspace admission or collision preflight), collision during execution (expected to be zero).
- **Graph:** stacked bars per configuration, showing what each stage of the pipeline contributes.

E12 Check resolution

- **Setup:** Thin-wall test with varying `collisionCheckStepDeg`.
- **Metrics:** missed collisions; preflight time.
- **Graph:** trade-off curve with two y-axes.

E13 Smoothness of motion

- **Setup:** One handover motion with the double-S profile against plain linear interpolation.
- **Metrics:** joint position, velocity, acceleration, and jerk.
- **Graph:** four stacked line plots.
- **Data:** the existing joint traces are sampled every 100 ms, which is too coarse for jerk; logging at every simulation step is required for this experiment.

E14 Simultaneous movement: region against exclusive arbitration

- **Metrics:** makespan; wait time per motion; number of concurrently moving arms; refused claims; minimum distance between arms.
- **Data:** already recorded in `concurrency.jsonl` and `live.jsonl`, together with the joint traces in the two `observed-*.jsonl` files.
- **Graph:** Gantt chart of the motions of each robot in both modes; cumulative distribution of wait times; minimum inter-arm distance over time with the clearance margin marked.

E15 Wall placement case study (optional)

- **Data:** `docs/wall_placement_and_space_lock.md`: the collision-free base rotation of Robot 3 increased from 120° to 360°, and the Robot 2–Robot 3 shared voxels from 30 to 132.
- **Graph:** before/after polar plot of the blocked base angles, illustrating the interplay of workspace estimation and collision checking.

6 End-to-end evaluation

E16 Scaling with the number of robots

- **Setup:** Two-, three-, four-, and five-robot scenes.
- **Metrics:** time per phase (neighbour discovery, interaction-space computation, cost pricing, route search, execution); number of messages; selected route.
- **Graph:** stacked bar chart of the phase durations per scene.

E17 Five-robot headline result

- **Metrics:** cumulative route costs (4.04 m via Robot 4 against 4.20 m via Robot 3); success rate, total time, and stability of the selected route over about ten runs.

- **Graph:** handover graph with edge costs and the selected route highlighted; table of the repeated runs.

E18 Rerouting (optional)

- **Setup:** Disable Robot 4 or block its shared region and verify that the cube is routed via Robot 3.
- **Note:** automatic re-planning after a failure (Stage 6 of `bellman_consensus_routing_plan.md`) is not implemented, so this is a restart with a modified configuration, not an in-run recovery.

7 Prioritization

Priority	Experiments	Rationale
Must-have	E3/E4, E6, E8, E11, E14, E17	One strong result per component plus the headline experiment.
Good	E1, E9, E10, E13, E16	Scaling and cost arguments; much of the data is already logged.
Nice-to-have	E2 visuals, E5, E7, E12, E15, E18	Additional depth and robustness.

E14 and E17 are the cheapest to produce, since their data is already logged. E3/E4 and E6 require small evaluation scripts but provide the most defensible numbers for the thesis's own methods.

Component coverage

Component	Experiments
Voxelization	E1, E2
Voxel size estimation	E3, E4, E5
Centrality (handover point selection)	E6, E7
Cheap neighbour approximation	E8, E9
Safety and movement	E10, E11, E12, E13, E14, E15
End-to-end (consensus and execution)	E16, E17, E18